Curvature Instability of an Active Gel Growing on a Wavy Membrane

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(Received 20 October 2025; accepted 29 April 2026; published 25 June 2026)

Cell shape changes are largely controlled by the actin cytoskeleton, a dynamic filament network beneath the plasma membrane. Several cell types can form extended freestanding protrusions not supported by an extracellular substrate or matrix, and regulated by proteins sensitive to the cell membrane curvature that modulate cytoskeletal dynamics. To gain insight into this process, we develop a theoretical model for the mechanics of a freestanding viscous actin network growing on a corrugated membrane. The model couples the dynamics of the viscous active gel with membrane deformation and the recruitment of curvature-sensitive actin nucleators. We show that an actin layer polymerising uniformly on the membrane always exerts a stabilizing effect that reduces membrane deformation. However, curvature-sensitive actin nucleator proteins can render the membrane linearly unstable, depending on the interplay between membrane and actin dynamics, giving rise to spontaneous membrane deformation which could initiate extended freestanding cellular protrusion.

DOI: 10.1103/xzkp-f581

Cell shape change is essential for a wide range of cellular functions, including migration and morphogenesis. It relies on the mechanical stress generated by the cytoskeleton on the cell membrane [1,2]. Actin polymerization at the membrane gives rise to dynamic structures such as lamellipodia, filopodia, and membrane ruffles [3,4]. At the edge of adherent cells (in the lamellipodium) actin polymerization against the membrane causes an actin retrograde flow, which experiences friction with the substrate mediated by adhesion proteins such as integrins, leading to a pushing force on the cell front [5,6]. Remarkably, many cell types such as immune cells can generate thin and extended (>10 μm long) freestanding pseudopods important for sensing and navigating through complex physiological environments [7]. Such actin-filled protrusions, which bear similarities with those observed in reconstituted *in vitro* actin/membrane systems [8], are not connected to any extracellular structures and must rely on purely internal stress within the actin gel in a way that is not understood.

Active gels growing on surfaces of constant curvature exert a compressive force on the surface whether growing on the convex or concave side of it [9–11]. This is a second order effect, hence not sensitive to the sign of the curvature and unlikely to trigger shape instability. Actin retrograde flow in adherent cells can trigger shape instability [12] and even symmetry breaking and waves for nonlinear adhesion forces [13], but this requires substrate friction. Here we study the stability of a freestanding actin gel growing on a deformable membrane using the theory of active gels [14]. The model, sketched in Fig. 1, includes the modulation

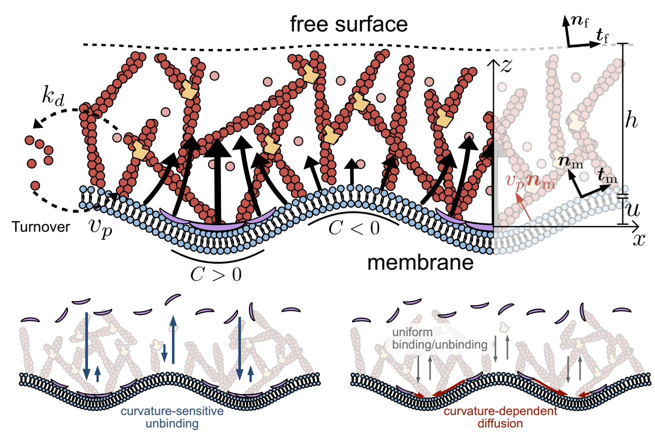

FIG. 1. Top: model for a viscous actin gel polymerizing on a wavy membrane. Balance between polymerization v_p at the membrane and depolymerization k_d yields a layer of finite thickness. Curvature sensitive proteins (purple) locally modulate actin polymerization. The location of the membrane ($u(x)$) and the local gel thickness ($h(x)$) are indicated, so are the corresponding normal and tangent vectors. Bottom: two possible scenarios of protein curvature sensing: a curvature-dependent unbinding rate (left) increases the protein bound time in regions of preferred curvature, and/or a spontaneous curvature (right) drives the diffusion of membrane-bound proteins toward regions of preferred curvature.

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of actin dynamics by proteins sensing the local membrane curvature. This is motivated by the fact that many curvature-sensitive proteins such as BAR-domain proteins can directly or indirectly regulate actin dynamics [15–18]. Such coupling between actin polymerization and curvature has been shown to give rise to instability and spontaneous large scale membrane deformation [19–21], but these models consider that actin polymerisation exerts a net force on the membrane, which is not appropriate for freestanding actin/membrane systems, in which the net force must vanish according to Newton's third law. Here we show that a gel with uniform polymerization always flattens the membrane, but that curvature modulation of actin dynamics can render the system linearly unstable, in a way that depends on the interplay between membrane and actin dynamics.

Model—Actin is polymerized at the membrane surface which creates a net flow of actin away from the membrane (Fig. 1). Actin turnover is included by assuming a constant depolymerization rate k_d throughout the actin layer and mass conservation is given by $\nabla\cdot(\rho_a \boldsymbol{v}) = -k_d\rho_a$, where ρ_a is the actin density and $\boldsymbol{v}$ the flow velocity. We assume here that the gel is incompressible (see Ref. [22] for the compressible case), such that

$$\nabla\cdot\boldsymbol{v} = -k_d. \tag{1}$$

The balance between polymerization and depolymerization ensures that the actin layer reaches a finite thickness denoted by $h(x)$ measured as the vertical distance from the membrane to the free surface. The growing actin layer is modelled as a viscous fluid described by the linear stress constitutive relation $\boldsymbol{\sigma} = -p\boldsymbol{I} + \eta(\nabla\boldsymbol{v} + \nabla\boldsymbol{v}^{\top})$, where p denotes the pressure and η the dynamic viscosity. Here, the pressure acts as a Lagrange multiplier ensuring constant density [cf. Eq. (1)]. At low Reynolds numbers and in the absence of external forces, momentum balance is given by $\nabla\cdot\boldsymbol{\sigma} = 0$, which, together with Eq. (1) leads to the Stokes equation

$$-\nabla p + \eta\nabla^2\boldsymbol{v} = 0. \tag{2}$$

We assume that the layer grows normal to the membrane, and that tangential flow is prohibited by strong friction with the membrane. The case of finite friction is discussed in Supplemental Material (SM) [23] and does not modify the normal stress for an incompressible gel. The boundary condition at the membrane, located at $u(x,t)$, is $\boldsymbol{v}|_{z=u(x)} = (v_p + \delta v_p(x) + \partial_t u/\sqrt{1+(\partial_x u)^2})\boldsymbol{n}_m(x)$, where $\boldsymbol{n}_m(x)$ denotes the unit normal vector at the membrane, v_p denotes a constant polymerization velocity and $\delta v_p(x)$ accounts for small, protein regulated local deviations from v_p. The free surface, located at $W(x) = u(x) + h(x)$, is stress-free (no normal or tangential stress) and its dynamics is given by: $\boldsymbol{v}\cdot\boldsymbol{n}_f = \partial_t W/\sqrt{1+(\partial_x W)^2}$.

In what follows, we assume small membrane deformations, i.e. $|\nabla u| \ll 1$ and neglect all terms of $\mathcal{O}(|\nabla u|^2)$ and higher. The membrane normal is then given by $\boldsymbol{n}_m \approx -\partial_x u(x)\boldsymbol{e}_x + \boldsymbol{e}_z$. The same strategy can be used for the free surface by replacing $u(x)$ by $W(x)$, as small deformations of the membrane surface lead to small deformations of the free surface $|\nabla h| \ll 1$. Consequently, all quantities may be expanded in independent Fourier modes, whose amplitudes are denoted by the subscript q, defined for the membrane shapes as: $u(x) = u_q e^{iqx}$. The validity of the linear regime is discussed in the SM [23].

Steady state actin layer—The stresses exerted by the polymerizing actin layer on the membrane is obtained by solving Eqs. (1), (2) for the velocity field in terms of Fourier modes (see details in SM [23].) The normal component reads,

$$\sigma_{nn,q} = -2\eta q(k_d u_q + \delta v_{p,q})\tanh(qv_p/k_d). \tag{3}$$

The variation of the normalized stress $\sigma_{nn,q}/2\eta k_d$ with qh_0 is shown in Fig. 2. An actin layer polymerizing uniformly ($\delta v_{p,q} = 0$) on a flat membrane ($u_q = 0$) does not generate any stress ($\sigma_{nn,q} = 0$) and reaches the steady-state thickness $h_0 = v_p/k_d$. For uniform polymerization on a wavy membrane shape ($u_q \neq 0$), the stress $\sigma_{nn,q}$ is positive for the

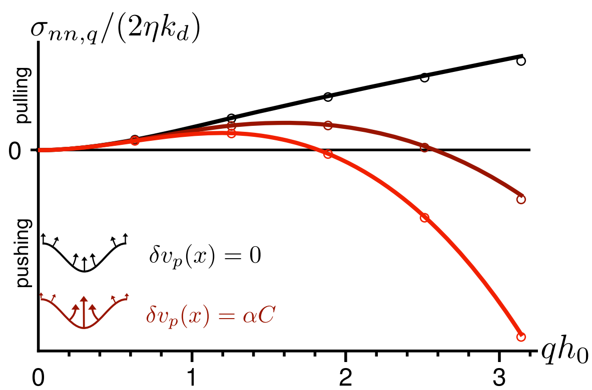

FIG. 2. Normal stress $\sigma_{nn,q}$ (Eq. (3)) exerted by a growing actin layer on a sinusoidal membrane as a function of the dimensionless layer thickness qh_0 for fixed dimensionless deformation $u_q/h_0 = 0.01$. For uniform polymerization (black curve) the stress is pushing on the peaks and pulling on the troughs (as shown here) of the wavy membrane, stabilizing the flat shape. It vanishes as $q \to 0$ and diverges linearly as $q \to \infty$. If the polymerization velocity varies linearly with the local curvature with a coupling strength α (red curves), the stress changes sign and destabilizes modes satisfying $q^2 h_0\alpha/v_p > 1$. The circles show the numerical results of a finite element method (see SM [23]). The parameters for the three curves are $\alpha/(h_0^2 k_d) = [0, 0.15, 0.3]$.

entire range of (nondimensional) layer thickness qh_0, indicating that the layer pushes on the peaks and pulls on the troughs of the sinusoidal shape, acting to reduce the membrane deformation. The stress vanishes as q^2 when $q \to 0$ and diverges as q when $q \to \infty$. Its dependence on the gel thickness $h_0 = v_p/k_d$ depends on whether the thickness is varied through the polymerization seed v_p or the depolymerization rate k_d. The stress from a thin layer ($h_0 \to 0$) vanishes if $v_p \to 0$ but remains finite if $k_d \to \infty$. The stress from a thick layer is finite if $v_p \to \infty$ but vanishes if $k_d \to 0$, consistent with previous results [8]. This is discussed further in SM [23].

The modulation of actin dynamics by curvature-sensitive proteins may be accounted for at the linear level by introducing a curvature-dependent modulation of the polymerization velocity of the form $\delta v_{p,q} = -\alpha q^2 u_q$, where α is the coupling strength. Inspection of Eq. (3) shows that the polymerization stress changes sign for $q > \sqrt{k_d/\alpha}$. This behavior is shown in Fig. 2 (blue line). The numerical solution of this system of equations using a finite element method (detailed in SM [23], code available on GitHub [42]) confirms these results and show excellent agreement with the linearized analytical solutions (Fig. 2).

Coupling actin and membrane dynamics—To understand how this mechanism can drive spontaneous membrane deformation, we derive the membrane shape dynamics by comparing the actin-generated forces to the membrane restoring forces [43] including the dynamics of curvature-sensitive proteins modulating actin dynamics. We consider a mixture of positively curved and negatively curved proteins, denoted by ρ_+ and ρ_-, with spontaneous curvature $\pm H$, respectively. We assume that the average protein coverage is the same for both protein types ($\int dS(\rho_+ - \rho_-) = 0$), so that the membrane under uniform protein coverage is flat. In this case, the protein density field relevant to membrane deformation only involves the signed protein density $\rho = \rho_+ - \rho_-$ in the limit of small deformation (see Ref. [44] and SM [23] for details). The membrane free energy includes an elastic contribution based on Helfrich free energy, a protein contribution written as a Ginzburg-Landau expansion and a coupling term due to protein spontaneous curvature [45–48]

$$\mathcal{F}[\rho, u] = \int d^2x \left[\frac{\kappa}{2}(\nabla^2 u)^2 + \frac{\gamma}{2}(\nabla u)^2 - \kappa\rho H \nabla^2 u + \frac{a}{2}\rho^2 + \frac{b}{2}(\nabla\rho)^2\right], \quad (4)$$

where κ and γ are the membrane bending rigidity and tension, and a and b are the strength of protein-protein interaction and cost of density gradients. From Eq. (4) we obtain the membrane restoring stress $\sigma_m = -\delta\mathcal{F}/\delta u = -\kappa\nabla^4 u + \gamma\nabla^2 u + \kappa H \nabla^2\rho$ and the generalized protein chemical potential $\mu = \delta\mathcal{F}/\delta\rho = -\kappa H \nabla^2 u + a\rho - b\nabla^2\rho$. The ratio $\sqrt{b/a}$ defines a length scale typically of order the protein size. As we are concerned with concentration modulation at (much) larger length scales, we neglect the concentration gradient energy term ($b = 0$).

We assume a linear relationship between actin polymerization modulation and the signed protein density with a coupling strength A:

$$\delta v_p = A\rho. \quad (5)$$

Demanding force balance between actin and membrane stresses, $\sigma_m + \sigma_{nn} = 0$, we obtain a dynamical equation for the membrane deformation. The time evolution of the signed membrane proteins density is governed by mass conservation $\partial_t\rho = \Lambda\nabla^2\mu + j_r$, where j_r is the recycling flux for the signed density, which may depend on curvature (see Fig. 1). The recycling flux for each protein type is written as $j_{r,\pm} = j_0/2 - k^0_{\mathrm{off}}(1 \mp \lambda_{\mathrm{off}} C)\rho_\pm$, such that proteins with positive spontaneous curvature unbind slower from regions of positive curvature [49], with a coupling strength λ_{off}. On a flat membrane, the average steady-state total protein density is $\rho_0 = j_0/k^0_{\mathrm{off}}$ and the recycling flux for the signed density is at lowest order $j_r = k^0_{\mathrm{off}}\rho_0(-\rho/\rho_0 + \lambda_{\mathrm{off}} C)$.

The equations are made dimensionless using the membrane mechanical length scale $\lambda = \sqrt{\kappa/\gamma}$ [6] as the characteristic length scale, λ/v_p as the timescale, and $2\eta v_p/\lambda$ as the stress scale. All resulting variables are expressed in dimensionless form and are denoted by a bar (e.g. $\bar{u}$, $\bar{\sigma}$, etc.; see Table S1 [?]). The dynamics of the system in Fourier space reads,

$$\partial_{\bar{t}} \begin{pmatrix} \bar{u}_q \\ \rho_q \end{pmatrix} = \begin{pmatrix} -\frac{1}{\bar{h}_0} - \frac{\bar{\gamma}(\bar{q}+\bar{q}^3)}{\tanh(\bar{q}\,\bar{h}_0)} & -\frac{\bar{\gamma}\,\bar{H}\,\bar{q}}{\tanh(\bar{q}\,\bar{h}_0)} - \bar{A} \\ -\bar{\Lambda}\,\bar{\gamma}\,\bar{H}\,\bar{q}^4 - \bar{k}^0_{\mathrm{off}}\bar{\lambda}_{\mathrm{off}}\rho_0\bar{q}^2 & -\bar{\Lambda}\,\bar{a}\,\bar{q}^2 - \bar{k}^0_{\mathrm{off}} \end{pmatrix} \begin{pmatrix} \bar{u}_q \\ \rho_q \end{pmatrix}. \quad (6)$$

As the trace of the dynamical matrix in Eq. (6) is always negative, the fixed point of the dynamical system is stable if the determinant of the matrix is positive, and turns into an unstable saddle point if Det < 0.

We first analyze the case of slow protein recycling and fast protein diffusion ($\Lambda \to \infty$, $k^0_{\mathrm{off}} \to 0$). Driven by the spontaneous curvature H, the protein density quickly adjusts to its steady-state distribution proportional to

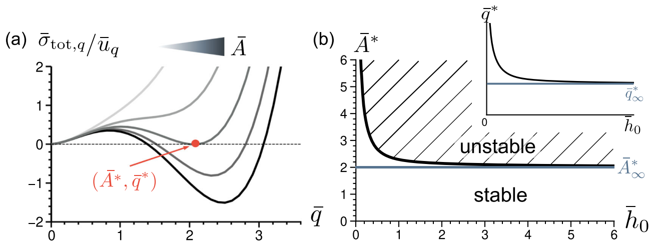

FIG. 3. Total steady-state membrane stress for fast protein diffusion without recycling. (a) Total normal stress as a function of the wave number $\bar{q}$ for different coupling strengths $\bar{A}$. A negative value indicates a destabilizing effect. (b) Critical coupling strength $\bar{A}^*$ at which the first unstable mode appears (wave vector q^*, red dot in panel (a)) as a function of the layer thickness. Inset shows the value of q^*. Parameters: $\bar{H} = 1$, $\bar{a} = 2$, $\bar{\gamma} = 1.5$ and $\bar{h}_0 = 10$ for panel (a).

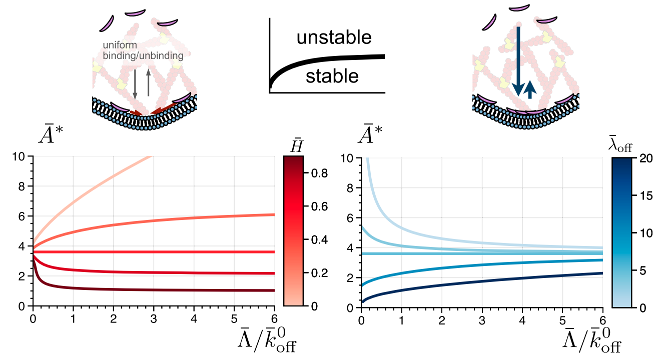

FIG. 4. Instabilities threshold A^* (the system is unstable if $A > A^*$) as a function of the ratio of protein mobility to recycling rate $\bar{\Lambda}/\bar{k}^0_{\mathrm{off}}$. One the left panel, the spontaneous curvature $\bar{H}$ is varied. at fixed curvature-sensitive unbinding rate $\bar{\lambda}_{\mathrm{off}} = 5$. On the right panel, $\bar{\lambda}_{\mathrm{off}}$ is varied at constant $\bar{H} = 0.5$. The asymptotic regimes of low and high mobility correspond to Eqs. (7) and (8). The instability is eased by protein mobility if $\bar{H} > \bar{a}\rho_0\bar{\lambda}_{\mathrm{off}}/\bar{\gamma}$ (see text). Parameters: $\bar{a} = 1$, $\bar{b} = 0$, $\bar{\gamma} = 1$, $\rho_0 = 0.1$, $\bar{h}_0 = 10$.

the local membrane curvature for any membrane shape: $\rho_q = -\bar{q}^2\bar{u}_q\bar{\gamma}\bar{H}/\bar{a}$. Even in the absence of protein/actin coupling ($A = 0$), this phenomenon is known to reduce the effective membrane bending rigidity, and can lead to the so-called "curvature instability" beyond a spontaneous curvature threshold, when the effective rigidity becomes negative and large membrane deformation spontaneously occurs [47]. Here we concentrate on the role of actin and remain below this threshold ($\bar{H} < \sqrt{\bar{a}/\bar{\gamma}}$ [47]). The total steady-state membrane stress is shown in Fig. 3(a) as a function of the wave number for different coupling strengths $\bar{A}$. For weak coupling ($\bar{A} \ll 1$), both the actin stress and the membrane stresses stabilize the flat membrane, so the stress remains of a given sign (positive—pulling—on the valleys of a sinusoidal shape). For strong enough coupling, the actin stress can destabilize a finite range of q modes. Since the membrane stress stabilizes large q modes, there exists a critical coupling strength $\bar{A}^*$ beyond which a range of modes around a critical wave mode q^* become unstable. This value is shown as a function of the layer thickness in Fig. 3(b), together with the corresponding value of q^*. Both are decreasing functions of the gel thickness, with asymptotic values for $q^*\bar{h}_0 \gg 1$:

$$\bar{A}^*_\infty = 2\frac{\bar{a}}{\bar{H}}\sqrt{1 - \bar{H}^2\bar{\gamma}/\bar{a}}, \quad \bar{q}^*_\infty = 1/\sqrt{1 - \bar{H}^2\bar{\gamma}/\bar{a}}. \tag{7}$$

We see that $A^*_\infty \to 0$ as $\bar{H} \to \sqrt{\bar{a}/\bar{\gamma}}$, consistent with the classical curvature instability discussed above [47].

In the opposite limit of very slow protein diffusion ($\Lambda \to 0$, $k^0_{\mathrm{off}} \to \infty$), the instability must rely only on curvature-dependent recycling, and one may show (see SM [23]) that the instability threshold (for thick layers) is given by

$$\bar{A}^*_0 = \frac{2\bar{\gamma}}{\rho_0\bar{\lambda}_{\mathrm{off}}}\sqrt{1 - \rho_0\bar{H}\bar{\lambda}_{\mathrm{off}}}, \quad \bar{q}^*_0 = 1/\sqrt{1 - \rho_0\bar{H}\bar{\lambda}_{\mathrm{off}}}. \tag{8}$$

This expression indicates a new type of actin-independent curvature instability, mediated by the curvature-dependent turnover of curvature-active proteins, which occurs if $\rho_0 H\lambda_{\mathrm{off}} > 1$ (see SM). Here again, we remain below this limit and concentrate on the role of actin.

The coupling threshold A^* is shown as a function of the ratio of in-plane diffusion to recycling rates $\bar{\Lambda}/\bar{k}^0_{\mathrm{off}}$ in Fig. 4 for the two curvature-sensing mechanisms displayed in Fig. 1. It varies monotonously between the asymptotes given by Eqs. (7) and (8). Therefore the instability is promoted by fast protein diffusion if protein localization by membrane curvature is more efficient via in-plane diffusion than via turnover, which corresponds to $\bar{A}^*_\infty < \bar{A}^*_0$, or $\bar{H} > \bar{a}\rho_0\bar{\lambda}_{\mathrm{off}}/\bar{\gamma}$. The left panel of Fig. 4 shows this behavior varying $\bar{H}$ at fixed $\bar{\lambda}_{\mathrm{off}}$ and the right panel varying $\bar{\lambda}_{\mathrm{off}}$ at fixed $\bar{H}$. As expected, the system is always stable ($A^* \to \infty$) for very slow diffusion in the absence of curvature-dependent recycling ($\bar{\lambda}_{\mathrm{off}} = 0$) and for very fast diffusion without spontaneous curvature ($\bar{H} = 0$).

Discussion—Our results may be summarized as follows: an actin gel growing on a corrugated membrane exerts a mechanical stress on the membrane. For an incompressible gel growing uniformly, the stress stabilizes flat membranes and can reach levels of order the elastic modulus of the gel $E \sim \eta k_d \simeq 10^3$ Pa for a gel of viscosity $\eta = 10^4$ Pa.s and of turnover rate $k_d = 0.1\ \mathrm{s}^{-1}$ [8]. Spontaneous actin-driven protrusion (linearly unstable) are predicted to occur in case actin dynamics (polymerisation rate) is sensitive to the curvature of the membrane it grows from. Curvature-sensing proteins are known to modulate actin polymerization [16,18]. Both N-BAR (Amphiphysin) and I-BAR (IRSp53, MTSS-1), respectively, attracted to negative and positive curvatures according to our convention, can promote actin recruitment [50,51]. Curvature sensing of

BAR domains involve both a spontaneous curvature and curvature-dependent recycling. We account for both effects (through the parameters $\bar{H}$ and $\bar{\lambda}_{\rm off}$, respectively) within a full linear model coupling actin and protein dynamics with membrane mechanics.

Our model shows the existence of an actin-driven curvature instability when proteins sensitive to positive curvature promote polymerization. This is in line with observations that I-BAR is implicated in protrusion formation [15,17]. This instability is predicted to occur if the value of the parameter $\bar{A}$, linearly coupling actin polymerization modulation to the curvature-dependent protein density (surface fraction) ρ according to $\delta v_p/v_p = \bar{A}\rho$, is beyond a threshold value A^*. Asymptotic expressions when either of the two curvature-sensing mechanisms dominates are given in Eqs. (7) and (8), and the full phase diagram is shown Fig. 4. Which mechanism dominates is largely controlled by the ratio of protein mobility to recycling rate. As the latter varies over a wide range for BAR proteins ($k^0_{\rm off}$: $0.01-10/{\rm sec}$ [49], corresponding to $\bar{\Lambda}/\bar{k}^0_{\rm off}$: $0.1-100$), this shows the relevance of exploring this parameter as done in Fig. 4. For appropriate physical parameters we find that $\bar{A}^* \simeq 5$–10, which corresponds to a doubling of actin polymerization speed in regions of favored curvature (with $\rho \simeq 10\%$). This seems like a reasonable threshold, reachable in cellular systems. The typical length scale of the first unstable mode is of order $2\pi/q^* \simeq 2\pi\lambda \simeq 500$ nm. It is comparable to the typical thickness of freestanding cellular protrusions (∼430–800 nm [7]). Although the latter certainly result from nonlinear coarsening effects beyond the linear stability analysis proposed here, such comparison is rather encouraging. Investigating this process further requires numerical resolution of the dynamical active gel equations near a deformable fluid surface in the highly non-linear regime. This technically challenging problem will be the goal of future studies, building on recent progress allowing to solve numerically the dynamics of a viscous fluid layer embedded in three-dimensional space [52].

Acknowledgments—This project has received funding from the European Research Council (ERC) ERC-SyG (Grant agreement ID: 101071793).

Data availability—The data that support the findings of this article are openly available [42].

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